

STOCHASTIC METHODS IN WATER RESOURCES

Unit 3: Random functions

Lecture 9: Definitions, random function types, stationary random functions

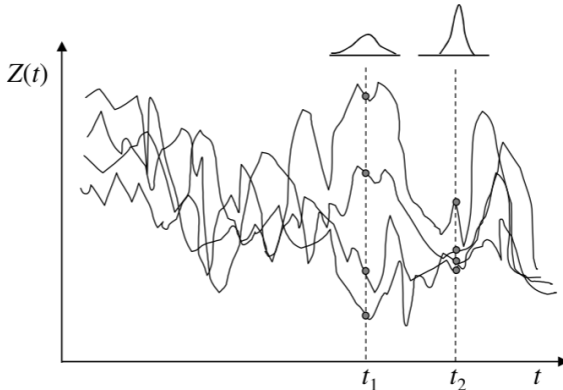
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September 25, 2025

Definitions

- ▶ Consider a **hydrological variable** Z that can either varies in space (e.g. hydraulic conductivity) or in time (e.g. surface water level) or in space and time (e.g. groundwater depth).
- ▶ The value of any of these variables is known only in those places where it is observed, and it is unknown in other places.
- ▶ Accordingly, the value of Z is uncertain, to deal with this, instead of using one single function either in time ($Z(t)$), or space ($Z(x)$) or space and time ($Z(x, t)$), multiple functions are used (see figure for $Z(t)$).
- ▶ Each of these functions have the same probability of representing the true.
- ▶ The family of equally probable functions representing Z is known as an **ensemble** or a **random function**.



Definitions

- ▶ **Random field**: If Z is a function of space ($\mathbf{x}$), $Z(\mathbf{x})$, $\mathbf{x} = (x, y, z)$.
- ▶ **Random or stochastic process**: If Z is a function of time (t), $Z(t)$.
- ▶ **Space-time random field**: If Z is a function of space ($\mathbf{x}$) and time (t), $Z(\mathbf{x}, t)$, $\mathbf{x} = (x, y, z)$.

Here, we are going to focus on **random process** ($Z(t)$). Note that in the figure above, four functions $Z(t)$ (**ensemble members**) are shown and each of them are known as **realizations** and denoted as $z(t)$. The number of **realizations**, depending on the variable's nature, can be finite or infinite.

Another way of explaining this concept is to imagine a **random function** as a collection of random variables (one at every location in space or point in time). Observing at the figure above, for each time t , a $Z(t)$ is defined. This is why for each t , $Z(t)$ is described with a **pdf** $f_Z(z; t)$, which depends not only on the value of z but also on the value of t . These **pdf** is estimated based on sample of data taken at particular time (e.g. t_1). As shown in the figure, the **pdfs** of the data sample at t_1 and t_2 are different, and indicate that variance is larger in t_1 than in t_2 and thus more uncertain. At each point in time (t), μ_Z is calculated as:

$$\mu_Z(t) = E[Z(t)] = \int_{-\infty}^{\infty} z f_Z(z; t) dz$$

and σ_Z^2 is:

$$\sigma_Z^2(t) = E[(Z(t) - \mu_Z(t))^2] = \int_{-\infty}^{\infty} (z - \mu_Z(t))^2 f_Z(z; t) dz$$

Definitions

The **random variables** are usually correlated in time (or in space for spatial random functions). This means that the **covariance** $\text{COV}[Z(t_1), Z(t_2)] \neq 0$. Note that the further apart the points (e.g. t_1 and t_2) the lower the **covariance**. This is estimated as

$$\begin{aligned}\text{COV}[Z(t_1), Z(t_2)] &= E[(Z(t_1) - \mu_Z(t_1))(Z(t_2) - \mu_Z(t_2))] \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (z_1 - \mu_Z(t_1))(z_2 - \mu_Z(t_2)) f_Z(z_1, z_2; t_1, t_2) dz_1 dz_2\end{aligned}$$

if $t_1 = t_2$, $\text{COV}[Z(t_1), Z(t_2)] = \sigma_Z^2(t)$.

For n different locations t_i where $i = 1, 2, \dots, n$, the probability measure that a realization of $Z(t)$ has the values between $z_1 < z(t_1) \leq z_1 + dz_1$,

$z_2 < z(t_2) \leq z_2 + dz_2, \dots, z_n < z(t_n) \leq z_n + dz_n$. The **associated multivariate pdf** $f_{Z(t_1), Z(t_2), \dots, Z(t_n)}(z_1, z_2, \dots, z_n; t_1, t_2, \dots, t_n)$

$$f_{Z(t_1), Z(t_2), \dots, Z(t_n)}(z_1, z_2, \dots, z_n; t_1, t_2, \dots, t_n) = \lim_{\partial z_1, \dots, \partial z_n \rightarrow 0} \frac{Pr(z_1 < z(t_1) \leq z_1 + \partial z_1, z_2 < z(t_2) \leq z_2 + \partial z_2, \dots, z_n < z(t_n) \leq z_n + \partial z_n)}{\partial z_1 \partial z_2 \dots \partial z_n}$$

The **multivariate pdf** of a **random function** is sometime referred to as **multipoint pdf**, because it relates to random variables at different points or locations. Note that **random function** is fully characterized if the **multivariate pdf** for any set of points is known.

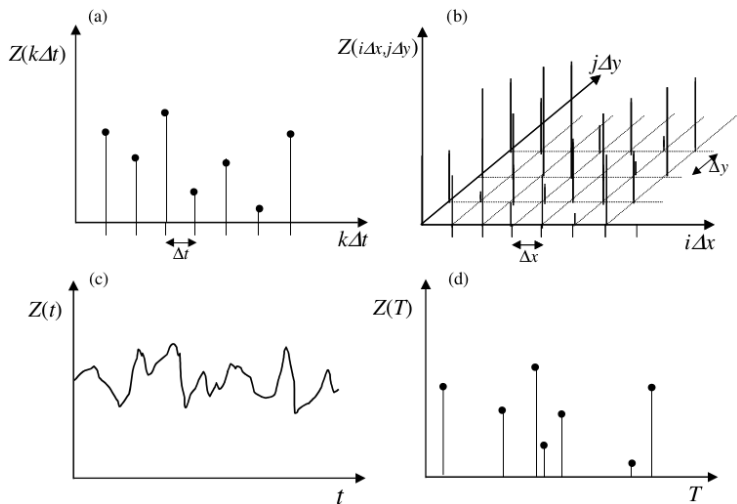
Types of random functions

Random functions can be classified as **continuous** whether the **random variable** is continuous (e.g. maximum annual river flows) or **discrete** whether the **random variable** is discrete (e.g. number of annual rainy days). Other way to classify a random function is based its domain (see figure). In the case of a **continuous random function**, this can be defined at:

1. **all locations** in time, space or space-time: $Z(t)$, $Z(\mathbf{x})$ or $Z(\mathbf{x}, t)$.
2. **discrete points** in time, space or space-time, where $Z(k\Delta t)$ ($k = 1, 2, \dots, K$) is referred to as a **random time series**, and $Z(i\Delta x, j\Delta y)$ ($i = 1, 2, \dots, I$; $j = 1, 2, \dots, J$) as a **lattice process**.
3. **random times** or **random locations** is space and time: $Z(T)$, $Z(\mathbf{X})$ or $Z(\mathbf{X}, T)$. This process is called **compound point process**. Note that the occurrence of the points at random coordinates in space and time is called a **point process**. If the occurrence of such a point is associated with the occurrence of a random variable (e.g. the occurrence of a thunder storm cell at a certain location in space with a random intensity of rainfall) it is called a compound point process.

The same distinction can be made for discrete-valued random functions: $D(t)$, $D(k\Delta t)$ or $D(T)$; $D(\mathbf{x})$, $D(i\Delta x, j\Delta y)$ or $D(\mathbf{X})$.

Types of random functions



This figure shows the different types of **random functions** based on the way they are defined on the functional domain: a) random time series, b) lattice process, c) continuous-time random function and d) compound point process.

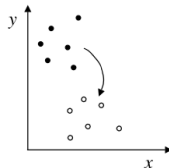
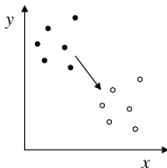
Stationary random functions

Strict stationary random functions

An special type of random functions are the **strict stationary random functions**. A **strict stationary random functions** is a random function for which its **multivariate pdf** is invariant under translation. This means that for any set of N locations and for any translations in time Δt , we have

$$f(z_1, z_2, \dots, z_N; t_1, t_2, \dots, t_N) = f(z_1, z_2, \dots, z_N; t_1 + \Delta t, t_2 + \Delta t, \dots, t_N + \Delta t) \quad \forall t_i, \Delta t$$

This assume that a configuration of points on the time axis can move backward and forward in time and the **multivariate pdf** does not change. For the spatial domain, **strict stationarity** means spatial no changes in the **pdf** if points are translated but no rotated. For instance a **strict random function** in 2D (see figure) the two sets of locations in the left-hand figure have the same **multivariate pdf**, while the locations in the right-hand figure necessarily does not. If the **multivariate pdf** of the **random function** is invariant under rotation, this is called a **statistically isotropic random function**.



Stationary random functions

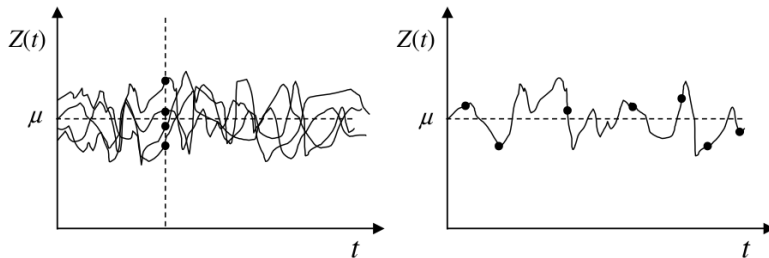
Ergodic random functions

- ▶ To estimate the statistical properties of a **random function**, a large number of realizations must be drawn.
- ▶ This is the case, for instance, of throwing a dice, from where one can generate multiple realizations.
- ▶ In reality this is not possible because we only have one realization: **reality**. This means that all relevant statistics of the **random function** are estimated from a single realization.
- ▶ In general, **ergodicity** deal with **stationarity** in **random functions**.
- ▶ To analyse **ergodicity**, statistical properties such as the **mean**, the **standard deviation** and the **covariance** are examined.
- ▶ **Stationarity** tells that all these properties are invariant no matter where you are (e.g. in time).
- ▶ For instance, for multiple realizations at point t_1 of $Z(t)$, the normal procedure to compute $\mu_Z(t_1)$ would be to average these realizations. However, this is impossible because in reality we only have one realization.
- ▶ If the **random function** is stationary the **pdf** $f_Z(z; t)$ at any location (in time) is the same and thus the statistical properties.
- ▶ This also means that within any single realisation we have at every location a sample from the same **pdf** $f_Z(z; t)$.

Stationary random functions

Ergodic random functions

- ▶ So, the **mean** can also be estimated if we take a sufficient number of samples from a single realisation, such as our **reality** (see figure below)



- ▶ This property of a **random function**, i.e. being able to estimate statistical properties of a random function from a large number of samples of a **single realisation** is called **ergodicity**.
- ▶ So for a **random function** is **ergodic** if it is **stationary**.
- ▶ Also for a **random function** to be **ergodic**, the samples from the single realisation should be taken from a large enough period of time or, in the spatial case, large enough area.

Stationary random functions

Ergodic random functions

A formal definition of **ergodicity** of a **random function** in its **mean**

$$\lim_{T \rightarrow \infty} \frac{1}{T} \int_T z(t) dt = \int_{-\infty}^{\infty} z f_Z(z; t) dz = \mu_Z$$

where T is an interval of time. This equation means that the integral over the probability distribution (the ensemble) for a given t (term on the left) can be replaced by a temporal (or spatial or spatio-temporal) integral of a very large interval T (or area or volume) (term on the right) to estimate μ_Z . volume).

Similarly, **ergodicity** of a **random function** in its **mean** is defined as

$$\begin{aligned} \lim_{T \rightarrow \infty} \frac{1}{T^2} \int_T z(t) z(t + \tau) dt &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (z_1 - \mu_Z)(z_2 - \mu_Z) f(z_1, z_2; t, t + \tau) dz_1 dz_2 \\ &= \text{COV}[Z(T), Z(t + \tau)] \quad \forall \tau \end{aligned}$$

where τ is delta time.

Stationary random functions

Second order stationary random functions

A weaker form of **stationarity** is second order **stationarity**. This means that the bivariate (or two-point) **probability distribution** is invariant under translation

$$f(z_1, z_2; t_1, t_2) = f(z_1, z_2; t_1 + \tau, t_2 + \tau) \quad \forall t_1, t_2, \tau$$

In practice, one require only **ergodicity** and thus **stationarity** for the **mean**, **varianza** and **covariance**. An **milder stationarity** is even considered and called **weak sense stationarity**. For **wide sense stationary random functions**

$$\mu_Z(t) = \mu_Z \quad \text{constant and } t \text{ or } x \text{ independent}$$

$$\sigma_Z(t) = \sigma_Z \quad \text{constant and } t \text{ or } x \text{ independent}$$

$$\text{COV}[Z(t_1)Z(t_2)] = C_Z(t_2 - t_1) = C_Z(\tau) \quad \text{depends on } \tau \text{ (separation in time or space)}$$

The **covariance function** is a graph that describe the covariance and depends on τ (**lag**). The covariance function start when $t_1 = t_2$ (variance of both variables) and when τ increase the function tends to zero. This means that a random variable is not correlated for larger τ .

In the case of **random space functions** the **weak sense stationarity** means that $\text{COV}[Z(\mathbf{x}_1), Z(\mathbf{x}_2)] = C_Z(\mathbf{x}_2 - \mathbf{x}_1) = C_Z(\mathbf{h})$, where $\mathbf{h} = \mathbf{x}_2 - \mathbf{x}_1$. In case of a **isotropic** random function $\text{COV}[Z(\mathbf{x}_1), Z(\mathbf{x}_2)] = C_Z(|\mathbf{x}_2 - \mathbf{x}_1|) = C_Z(|\mathbf{h}|) = C_Z(h)$, where $|\mathbf{h}| = h$ is the **norm** of the difference vector.

Stationary random functions

Second order stationary random functions

When the **covariance function** is divided by the **variance** σ_Z^2 , this becomes the **correlation function** either $\rho_Z(\tau) = C_Z(\tau)/\sigma_Z^2$ or $\rho_Z(\mathbf{h}) = C_Z(\mathbf{h})/\sigma_Z^2$. When the data is regularly spaced either in time or space, the **covariance function** is:

$$\hat{C}_Z(k\Delta t) = \frac{1}{n-k} \sum_{k=1}^{n-k} (z_i - \hat{\mu}_Z)(z_{i+k} - \hat{\mu}_Z)$$

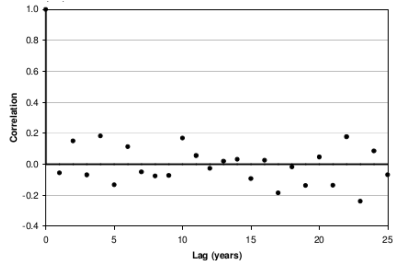
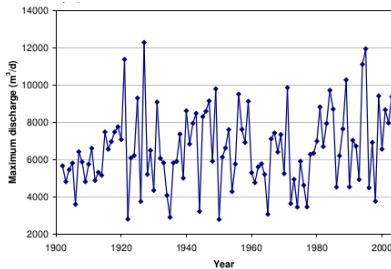
where $k\Delta t$ is the lag (in units of time or space) and Δt is the distance (in time or space) between observations. Frequently the estimator $\hat{C}_Z(k\Delta t)$ is used in the analysis of time series. In the case of irregularly positioned data in space, the **covariance function** is

$$\hat{C}_Z(\mathbf{h}) = \frac{1}{n(\mathbf{h})} \sum_{i=1}^{n(\mathbf{h})} (z(\mathbf{x}_i) - \hat{\mu}_Z)(z(\mathbf{x}_i + \mathbf{h} \pm \Delta \mathbf{h}) - \hat{\mu}_Z)$$

where $\mathbf{h}$ is the lag (a vector in space), $\Delta \mathbf{h}$ is a lag-tolerance which is needed to group a number of data-pairs together to get stable estimates for a given lag and $n(\mathbf{h})$ are the number of data pairs that are a distance $\mathbf{h} \pm \Delta \mathbf{h}$ apart.

Stationary random functions

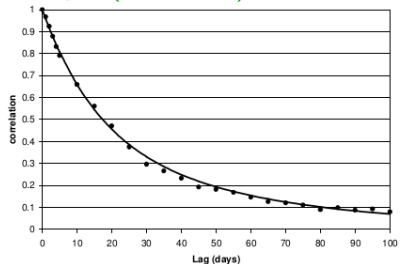
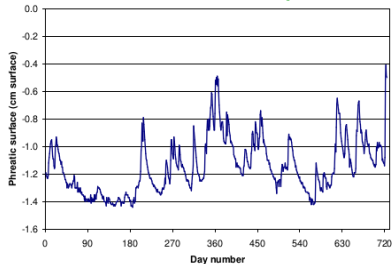
Correlation function: maximum annual streamflow



- ▶ The correlation plot, shows the **estimated** and the **fitted** correlation functions. According to these plots the data is uncorrelated.
- ▶ This uncorrelated behaviour of lagged data points is a characteristics of **white noise**, which are pure random processes, where the data points are uncorrelated no matter τ .
- ▶ **White noise** follow a **normal distribution**, so that, one can produce synthetic e.g. annual maximum streamflows after sampling the process. It would produce a time series with the same statistical properties of the original one.

Stationary random functions

Correlation function: daily water table depth (1994-1995)



- ▶ The correlation plot, shows the **estimated** and the **fitted** correlation functions. According to these plots the correlation decrease exponentially as τ increase.
- ▶ Again, one can produce synthetic e.g. daily water levels after sampling the process at the same discrete times. It would produce a time series with the same statistical properties of the original one.

Stationary random functions

Second order stationary random functions

The following table shows some models for modelling the **covariance function** for **isotropic random functions** and **wide sense stationary process**.

Exponential covariance	$C_Z(h) = \sigma_Z^2 e^{-h/a}$ for $h \geq 0, a > 0$
Gaussian covariance	$C_Z(h) = \sigma_Z^2 e^{-h^2/a^2}$ for $h \geq 0, a > 0$
Spherical covariance	$C_Z(h) = \begin{cases} \sigma_Z^2 \left[1 - \frac{3}{2} \left(\frac{h}{a} \right) + \frac{1}{2} \left(\frac{h}{a} \right)^3 \right], & \text{if } h < a \\ 0, & \text{if } h \geq a, h \geq 0, a > 0 \end{cases}$
Hole effect (wave) model	$C_Z(h) = \sigma_Z^2 \frac{b \sin(h/a)}{h}$ for $h \geq 0, a > 0$
White noise model	$\rho_Z(h) = \begin{cases} 1, & \text{if } h = 0 \\ 0, & \text{if } h > 0 \end{cases}$

Note that the **white noise model** has infinity variance, so it is not **wide sense stationary**.

Relations between various forms of stationarity

If a **random function** is **wide sense stationary** and its **multivariate pdf** is a **Gaussian distribution**, It is also second order stationary and also a strict sense **stationary random function**. More important, a wide sense stationary random function that is multivariate Gaussian (and thus also strict stationary) is completely characterised by only a few statistics: a constant **mean** $\mu_Z(t) = \mu_Z$ and a **covariance function** $C_Z(t_2 - t_1)$ that only depends on the separation distance.

Stationary random functions

Intrinsic random functions

For an intrinsic random function we require (we show the spatial form here)

$$E[Z(\mathbf{x}_2) - Z(\mathbf{x}_1)] = 0 \quad \forall \mathbf{x}_2, \mathbf{x}_1$$